Arith Maldonado Zamudio

Biomedical Engineer | Mechatronics Engineering Student | Mechanical Design, CAD & Product Development

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GPA: 9.1/10.0 | Portfolio: aritmaldzamu.github.io/portfolio-ingenieria-mecanica

PROFILE

Mechatronics Engineering student and graduated Biomedical Engineer (Honors Mention), specialized in mechanical design, product development, and R&D in advanced manufacturing environments. Hands-on experience with 3D CAD modeling (SolidWorks, Creo Parametric), FEA simulation, engineering drawings, assemblies, and BOMs; complemented by practical manufacturing skills (CNC, lathe, milling, laser cutting, 3D printing). Capable of applying DFM principles to translate design concepts into functional prototypes and collaborate with cross-functional engineering teams to solve product-related challenges.

EDUCATION

Universidad Iberoamericana Puebla — *Expected Dec. 2026*

B.Eng. Mechatronics Engineering (in progress) — *Puebla, Mexico*

GPA: 9.1/10.0 | Relevant coursework: Strength of Materials, Advanced Control Systems, Embedded Systems, Mechanical Design, and Industrial Automation.

Universidad Iberoamericana Puebla — *2020 - 2024*

B.Eng. Biomedical Engineering, Graduated with Honors Mention — *Puebla, Mexico*

GPA: 9.1/10.0 | Focus: biomechanics, medical device design, finite element analysis, and prototyping.

RELEVANT EXPERIENCE

Medical Equipment Technical Specialist & Applications Specialist — *Jul. 2024 - Dec. 2024*

Punto Focal Equipo Médico — *Puebla, Mexico*

- Preventive/corrective maintenance, installation, and diagnosis of ultrasound equipment, X-ray systems, flat panels, and triggering devices.
- Managed 15+ ultrasound-related cases: technical reports, client follow-up, warranty/ticket processing, and basic diagnostics.
- Delivered technical training and product demonstrations to physicians, technicians, and clients.
- Identified a functional replacement for a medical power supply, reducing the client cost from approx. MXN $15,000 to MXN $1,500 - applying cost-driven design evaluation.

SELECTED PROJECTS

3-DOF Ball Balancing Platform — *Mechatronics*

ESP32, SG90 servos, Python, OpenCV, PID control, Bluetooth communication, SolidWorks (structural design)

- Designed and manufactured the mechanical platform structure; generated assembly drawings and part specifications to support fabrication.
- Implemented closed-loop control integrating servo actuation, camera feedback (OpenCV), and continuous PID - demonstrating full hardware-software integration.

XGIO - GPS Ecosystem for Smart Cane — *Social Service*

Mobile app, backend, admin dashboard, Firebase, Vercel, CAD/SolidWorks documentation

- Developed the full GPS tracking system (app, backend, admin panel, user-device registration) for visually impaired individuals; produced CAD documentation and technical reports.

AWARDS & CERTIFICATIONS

- Poster accepted at ISPO 20th World Congress 2025, Stockholm: "Redesign and validation of a tool holding system for a transradial prosthesis for a stomatology student"; Honors Mention in Biomedical Engineering, Dec. 2024.
- CSWA (Certified SOLIDWORKS Associate); Certified SOLIDWORKS Simulation Associate; Certified SOLIDWORKS Additive Manufacturing Associate; Google Project Management Certificate; Siemens Basics of Robotics; Siemens Expedite - Skills for Industry: Industry Foundations; EF SET English B2; German A2.

SKILLS

CAD / FEA:	SolidWorks (CSWA certified), Creo Parametric, SolidWorks Simulation, engineering drawings, assemblies, BOMs, GD&T, metrology, and CAD documentation.
Product Development:	DFM principles, design reviews, rapid prototyping, tolerance analysis, and product lifecycle documentation.
Manufacturing:	CNC / router CNC, lathe, milling machine, laser cutting, 3D printing, and blueprint interpretation.
Programming & Analysis:	Python, OpenCV, MATLAB, Arduino/ESP32, Power BI, Excel, Firebase, and technical reporting.
Languages:	Native Spanish; Advanced English (B2); German A2.
Availability:	Full-time engineering roles from July 2026; interest in mechanical design, product development, advanced manufacturing, and electromechanical systems.